A Multifactor Perspective on Volatility-Managed Portfolios

Victor DeMiguel   Alberto Martín-Utrera   Raman Uppal*

*Victor DeMiguel is at London Business School. Alberto Martín-Utrera is at Iowa State University. Raman Uppal is at EDHEC Business School and CEPR.

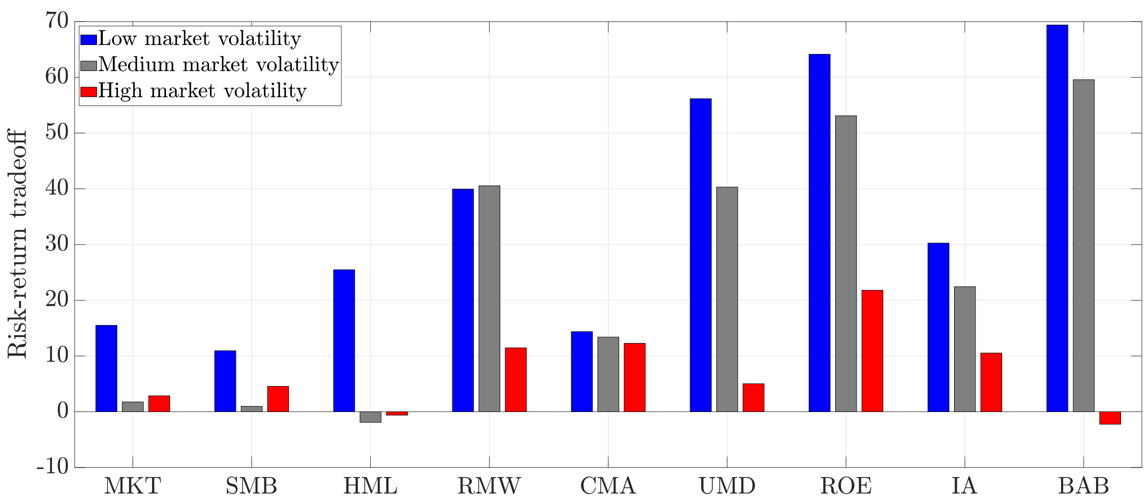
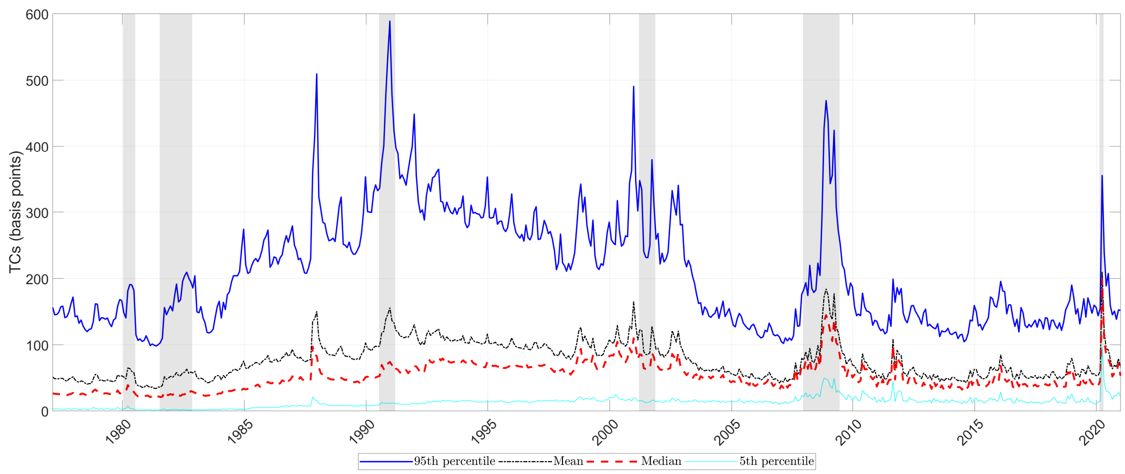


Figure 1. Factor risk-return tradeoff and market volatility. This barplot illustrates how the risk-return tradeoff for the nine factors in our dataset varies with realized market volatility. We first use the monthly time series of realized market volatility to sort the months in our sample into terciles. For each factor, we then estimate the risk-return tradeoff for month t as the realized factor return for month $t+1$ divided by the monthly realized factor variance estimated as the sample variance of daily returns for month t. Finally, we report the risk-return tradeoff averaged across the months in each tercile. Blue bars correspond to the tercile containing low-market-volatility months, gray bars to medium-market-volatility months, and red bars to high-market-volatility months. The sample spans January 1977 to December 2020.

Panel A. Individual-stock level



Panel B. Factor level

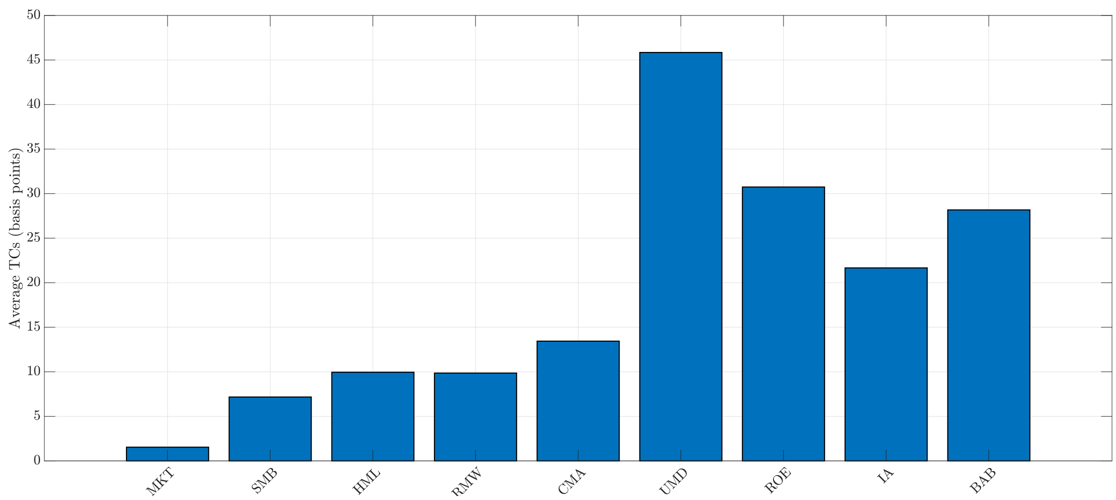


Figure 2. Transaction costs at the stock and factor levels. This figure depicts several descriptive statistics of the transaction costs estimated using the method of **?** described in Equation (**??**). Panel A depicts how the 95th percentile, mean, median, and 5th percentile of individual-stock transaction-cost parameters vary over time for the out-of-sample period from January 1977 to December 2020, with NBER recessions shaded in gray. Panel B depicts the average monthly transaction cost of trading the nine factors in our dataset.

Table I
Performance of Volatility-managed Individual-factor Portfolios

For each of the nine factors we consider, this table reports the annualized Sharpe ratios of the unmanaged factor, $\mathrm{SR}(r_k)$, the volatility-managed individual-factor portfolio, $\mathrm{SR}(r_k, r_k^\sigma)$, which is the mean-variance combination of the unmanaged factor with its managed counterpart given in (**??**), and the p-value for the difference between these two Sharpe ratios. We consider an investor with risk-aversion parameter $\gamma = 5$. Panel A reports performance in-sample and ignoring transaction costs, Panel B in-sample and net of costs but ignoring trading diversification, Panel C out-of-sample but ignoring costs, Panel D out-of-sample and net of costs but ignoring trading diversification, and Panel E out-of-sample and net of costs considering trading diversification. To facilitate comparison, both the in-sample and out-of-sample performance are evaluated for January 1977 to December 2020.

	MKT	SMB	HML	RMW	CMA	UMD	ROE	IA	BAB
Panel A: In-sample without transaction costs									
$\mathrm{SR}(r_k)$	0.530	0.208	0.170	0.506	0.399	0.474	0.722	0.508	0.880
$\mathrm{SR}(r_k, r_k^\sigma)$	0.585	0.246	0.215	0.739	0.419	1.088	1.153	0.621	1.397
p-value($\mathrm{SR}(r_k, r_k^\sigma)$-$\mathrm{SR}(r_k)$)	0.245	0.369	0.335	0.037	0.313	0.000	0.001	0.098	0.000
Panel B: In-sample net of transaction costs but without trading diversification									
$\mathrm{SR}(r_k)$	0.519	0.125	0.053	0.356	0.159	0.114	0.311	0.107	0.606
$\mathrm{SR}(r_k, r_k^\sigma)$	0.521	0.125	0.053	0.356	0.159	0.251	0.331	0.107	0.703
p-value($\mathrm{SR}(r_k, r_k^\sigma)$-$\mathrm{SR}(r_k)$)	0.457	0.500	0.500	0.500	0.500	0.242	0.407	0.500	0.160
Panel C: Out-of-sample without transaction costs									
$\mathrm{SR}(r_k)$	0.530	0.208	0.170	0.506	0.399	0.474	0.722	0.508	0.880
$\mathrm{SR}(r_k, r_k^\sigma)$	0.408	0.068	0.194	0.527	0.355	1.035	1.094	0.605	1.321
p-value($\mathrm{SR}(r_k, r_k^\sigma)$-$\mathrm{SR}(r_k)$)	0.894	0.927	0.389	0.465	0.894	0.000	0.001	0.063	0.000
Panel D: Out-of-sample net of transaction costs but without trading diversification									
$\mathrm{SR}(r_k)$	0.519	0.125	0.053	0.356	0.159	0.114	0.311	0.107	0.606
$\mathrm{SR}(r_k, r_k^\sigma)$	0.324	−0.295	−0.041	−0.453	−0.047	0.194	0.269	−0.127	0.690
p-value($\mathrm{SR}(r_k, r_k^\sigma)$-$\mathrm{SR}(r_k)$)	0.976	1.000	0.876	1.000	1.000	0.337	0.676	0.999	0.279
Panel E: Out-of-sample net of transaction costs with trading diversification									
$\mathrm{SR}(r_k)$	0.519	0.125	0.053	0.356	0.159	0.114	0.311	0.107	0.606
$\mathrm{SR}(r_k, r_k^\sigma)$	0.433	0.035	0.089	0.226	0.153	0.209	0.324	0.193	0.746
p-value($\mathrm{SR}(r_k, r_k^\sigma)$-$\mathrm{SR}(r_k)$)	0.922	0.856	0.243	0.960	0.543	0.100	0.407	0.025	0.064

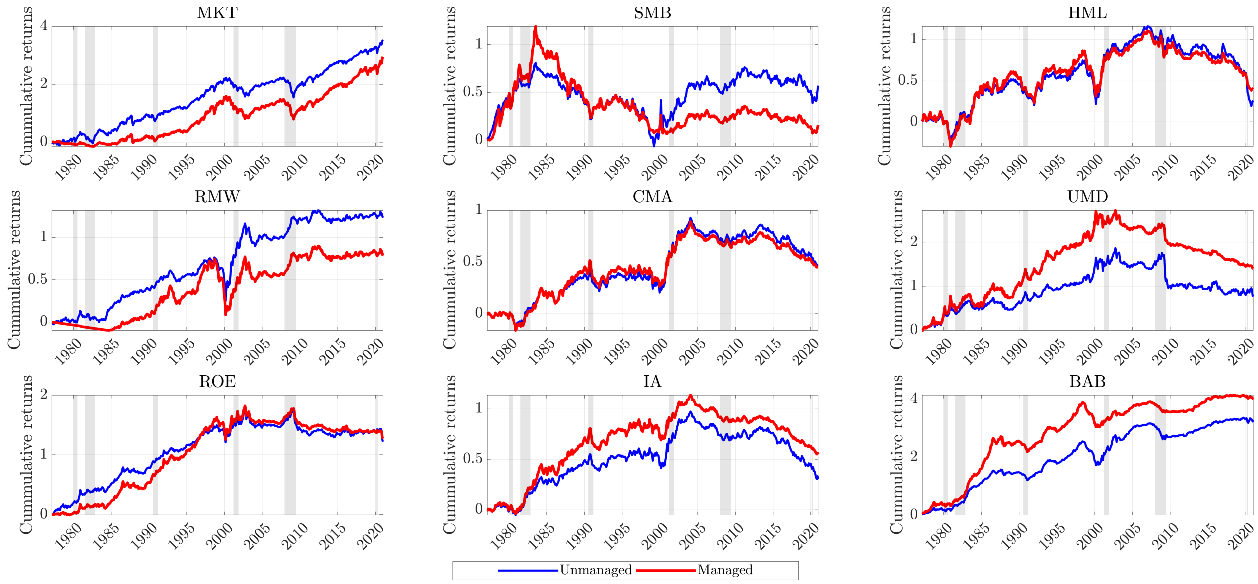


Figure 3. Cumulative returns of individual-factor portfolios. The nine graphs in this figure depict the out-of-sample cumulative returns net of transaction costs with trading diversification of each unmanaged factor (blue line) and its associated volatility-managed individual factor portfolio (red line) over the out-of-sample period from January 1977 to December 2020. The cumulative returns are reported in dollars, and the volatility-managed individual factor portfolio is scaled to have the same volatility as the unmanaged factor.

Table II
Performance of Conditional Mean-variance Multifactor Portfolio

This table reports the out-of-sample and net-of-costs performance of two multifactor portfolios: the conditional mean-variance multifactor portfolio (CMV) obtained by solving problem (**??**) and the unconditional mean-variance multifactor portfolio (UMV) obtained by solving problem (**??**) under the additional constraint that $b_k = 0$ for $k = 1, 2, \ldots, K$; that is, under the constraint that its weights on the K factors are constant over time. For each multifactor portfolio, the table reports the out-of-sample annualized mean, standard deviation, Sharpe ratio of returns net of transaction costs accounting for trading diversification, and p-value for the difference between the Sharpe ratios of the conditional and unconditional portfolios. The table also reports the annualized alpha of the time-series regression of the conditional portfolio out-of-sample returns net of transaction costs on those of the unconditional portfolio, alpha Newey-West t-statistic, and out-of-sample transaction costs of the unconditional and conditional portfolios. The portfolios are constructed using all nine factors in our dataset. We report out-of-sample performance from January 1977 to December 2020.

	UMV	CMV
Mean	0.430	0.477
Standard deviation	0.458	0.449
Sharpe ratio	0.940	1.062
p-value(SR_{CMV}-SR_{UMV})		0.006
α		0.066
$t(\alpha)$		3.638
TC	0.163	0.213

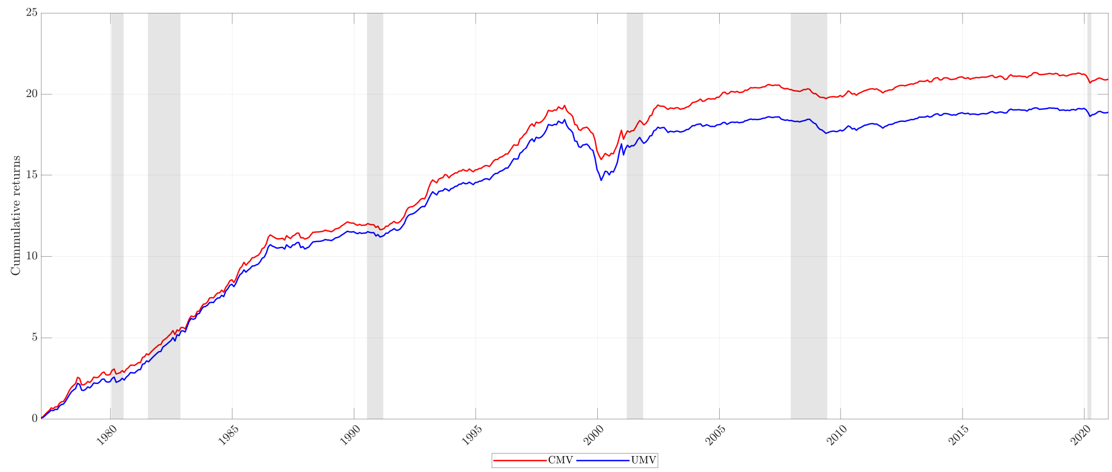


Figure 4. Cumulative returns of multifactor portfolios. This figure depicts the out-of-sample cumulative returns net of transaction costs of the unconditional (UMV) and conditional (CMV) mean-variance multifactor portfolios over the out-of-sample period from January 1977 to December 2020. The returns are reported in dollars and the conditional multifactor portfolio is standardized to have the same volatility as its unconditional counterpart.

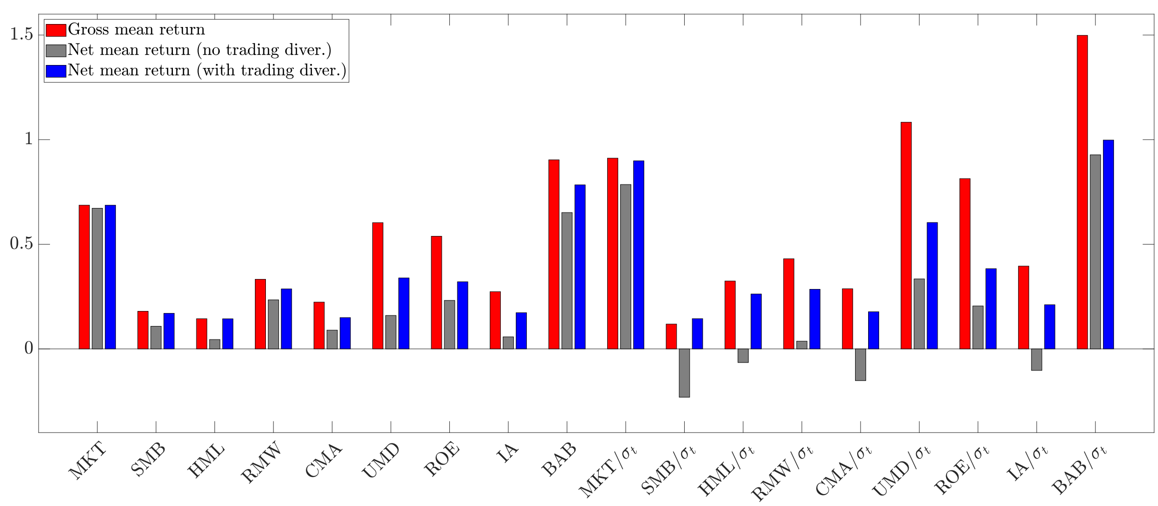


Figure 5. Gross and net-of-costs mean factor returns. This barplot depicts the monthly average factor returns (in percentage) of the nine unmanaged and volatility-managed factors. The figure compares three quantities for each factor: (i) its mean gross return, (ii) its mean return net of transaction costs ignoring trading diversification, and (iii) its mean return net of transaction costs accounting for trading diversification. For the case in which we account for trading diversification, we use the factor weights that solve problem (**??**) in sample. The sample spans January 1977 to December 2020.

Table III
Understanding the Performance of the Multifactor Portfolios

This table reports the performance of three multifactor portfolios optimized either ignoring or accounting for transaction costs. The three multifactor portfolios are the: (i) unconditional multifactor portfolio (UMV), (ii) conditional fixed-weight multifactor portfolio (CFW) of **?**, and (iii) our conditional multifactor portfolio (CMV). Columns (1) to (3) report the performance of the three multifactor portfolios optimized ignoring transaction costs and Columns (4) to (6) accounting for transaction costs. Each of the four panels reports the performance of the portfolios *evaluated* in a different way: Panel A in-sample without transaction costs, Panel B out-of-sample without transaction costs, Panel C out-of-sample with transaction costs without trading diversification, and Panel D out-of-sample with transaction costs with trading diversification. The sample period and quantities reported for each portfolio are the same as in Table **??**.

	Optimized ignoring TC			Optimized accounting for TC		
	UMV (1)	CFW (2)	CMV (3)	UMV (4)	CFW (5)	CMV (6)
Panel A: In-sample without transaction costs						
Mean	0.415	0.602	0.680	0.300	0.314	0.432
Standard deviation	0.288	0.347	0.369	0.218	0.222	0.250
Sharpe ratio	1.441	1.735	1.844	1.378	1.415	1.726
p-value(SR_{CMV}-SR_{UMV})		0.000	0.000		0.005	0.000
α		0.187	0.213		0.009	0.107
$t(\alpha)$		5.738	6.684		3.073	7.638
TC	0.000	0.000	0.000	0.000	0.000	0.000
Panel B: Out-of-sample without transaction costs						
Mean	0.753	0.783	0.925	0.593	0.626	0.690
Standard deviation	0.580	0.520	0.569	0.458	0.445	0.449
Sharpe ratio	1.299	1.506	1.625	1.295	1.407	1.537
p-value(SR_{CMV}-SR_{UMV})		0.000	0.000		0.001	0.000
α		0.140	0.239		0.059	0.125
$t(\alpha)$		5.012	5.797		3.809	6.415
TC	0.000	0.000	0.000	0.000	0.000	0.000
Panel C: Out-of-sample with transaction costs without trading diversification						
Mean	0.411	0.313	0.349	0.347	0.343	0.332
Standard deviation	0.580	0.520	0.569	0.458	0.445	0.449
Sharpe ratio	0.710	0.601	0.613	0.758	0.772	0.739
p-value(SR_{CMV}-SR_{UMV})		0.984	0.935		0.333	0.728
α		-0.035	-0.027		0.012	0.001
$t(\alpha)$		-1.610	-0.727		0.967	0.032
TC	0.341	0.470	0.576	0.246	0.283	0.358
Panel D: Out-of-sample with transaction costs with trading diversification						
Mean	0.517	0.511	0.575	0.430	0.457	0.477
Standard deviation	0.580	0.520	0.569	0.458	0.445	0.449
Sharpe ratio	0.891	0.984	1.010	0.940	1.026	1.062
p-value(SR_{CMV}-SR_{UMV})		0.017	0.078		0.003	0.005
α		0.072	0.103		0.046	0.066
$t(\alpha)$		3.094	2.759		3.408	3.638
TC	0.236	0.272	0.350	0.163	0.169	0.213

Table IV
Sources of Trading-diversification Benefits

This table reports the out-of-sample and net-of-costs performance of the unconditional (UMV) and conditional (CMV) multifactor portfolios. We evaluate the performance of the conditional multifactor portfolio for three different cases: (1) taking trading diversification fully into account (that is, netting trades across all unmanaged and managed factors), (2) taking trading diversification into account only partially (netting trades only across the unmanaged and managed versions of each individual factor, but not across different factors), and (3) ignoring trading diversification altogether. The sample period and quantities reported for each portfolio are the same as in Table **??**.

	(1) with full trading div. within & across factors		(2) with trading div. only within factors	(3) without any trading div.
	UMV	CMV	CMV	CMV
Mean	0.430	0.477	0.351	0.332
Standard deviation	0.458	0.449	0.449	0.449
Sharpe ratio	0.940	1.062	0.783	0.739
p-value(SR_{CMV}-SR_{UMV})		0.005	1.000	1.000
α		0.066	−0.060	−0.079
$t(\alpha)$		3.638	−3.227	−4.223
TC	0.163	0.213	0.339	0.358

Table V
Factor Risk Prices and Market Volatility

This table reports the coefficients α_k and β_k for the time-series regression defined in Equation (**??**) for the nine factors in our dataset. The numbers in square brackets are Newey-West t-statistics. The sample spans January 1977 to December 2020.

	MKT	SMB	HML	RMW	CMA	UMD	ROE	IA	BAB
α_k	6.668	5.479	7.573	30.467	13.341	33.565	46.118	20.979	41.824
	[3.790]	[1.494]	[1.258]	[4.499]	[2.240]	[6.676]	[7.101]	[3.820]	[8.678]
β_k	0.512	0.155	1.002	1.155	0.329	1.561	1.361	0.770	2.311
	[2.358]	[0.420]	[2.270]	[2.236]	[0.749]	[3.840]	[2.621]	[1.982]	[7.204]

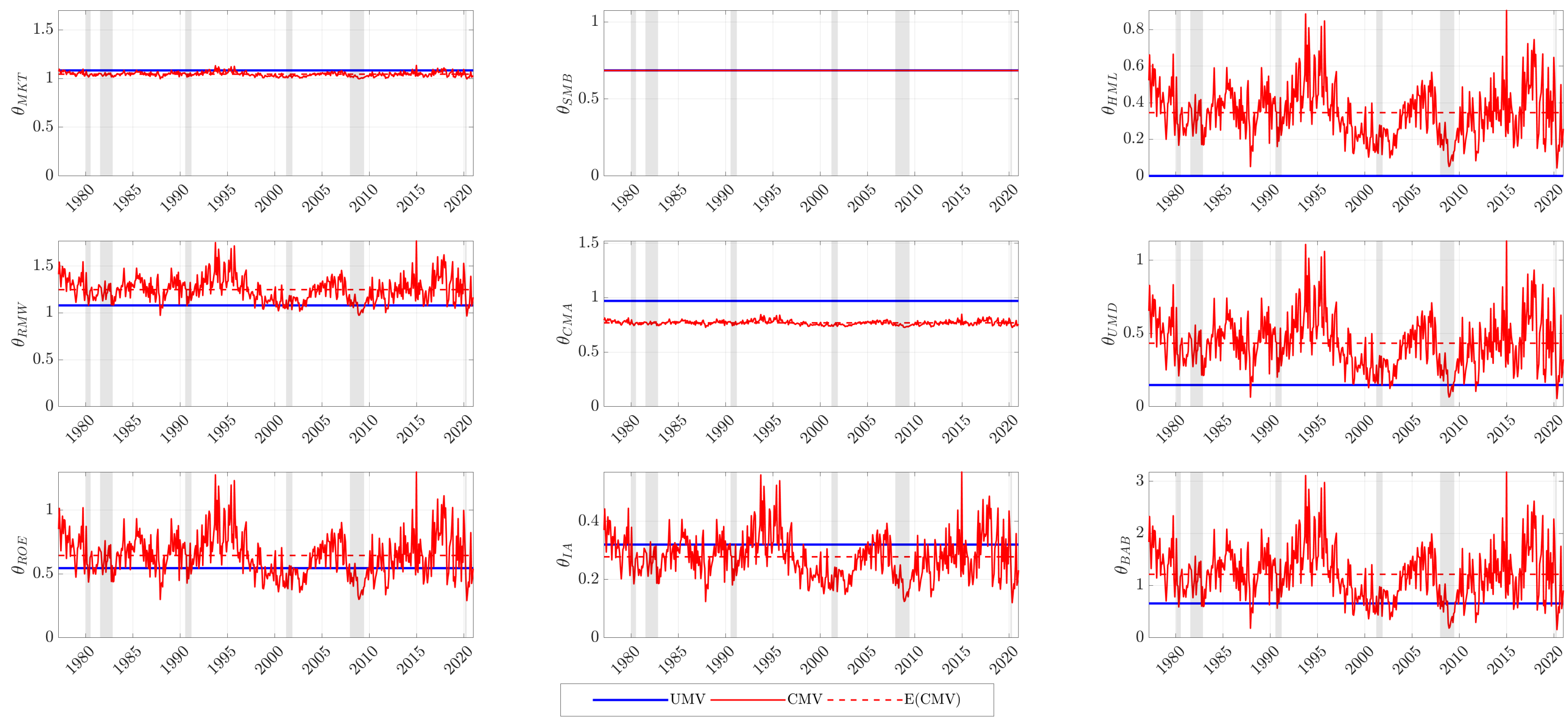


Figure 6. Weights of unconditional and conditional multifactor portfolios. This figure depicts the in-sample weights of the unconditional multifactor portfolio (UMV, blue line) and the conditional multifactor portfolio (CMV, solid red line) from January 1977 to December 2020. The figure also depicts the average weights of the conditional multifactor portfolio (E[CMV], dashed red line). Each of the nine graphs depicts the weights for a particular factor.